Chapter 13

Classification of Covering Spaces

Up to this point, we have used covering spaces primarily as a tool for computing fundamental groups. Now we turn things around and use the fundamental group as a tool for studying covering spaces.

To do this in any reasonable way, we must restrict ourselves to the case where B is locally path connected. Once we have done this, we may as well require B to be path connected as well, since B breaks up into the disjoint open sets B_α that are its path components, and the maps $p^{-1}(B_\alpha) \to B_\alpha$ obtained by restricting p are covering maps, by Theorem 53.2. We may as well assume also that E is path connected. For if E_α is a path component of $p^{-1}(B_\alpha)$, then the map $E_\alpha \to B_\alpha$ obtained by restricting p is also a covering map. (See Lemma 80.1.) Therefore, one can determine all coverings of the locally path-connected space B merely by determining all path-connected coverings of each path component of B!

For this reason, we make the following:

Convention. *Throughout this chapter, the statement that $p : E \to B$ is a covering map will include the assumption that E and B are locally path connected and path connected, unless specifically stated otherwise.*

With this convention, we now describe the connection between covering spaces of B and the fundamental group of B.

If $p : E \to B$ is a covering map, with $p(e_0) = b_0$, then the induced homomorphism p_* is injective, by Theorem 54.6, so that

$$H_0 = p_*(\pi_1(E, e_0))$$

is a subgroup of $\pi_1(B, b_0)$ isomorphic to $\pi_1(E, e_0)$. It turns out that the subgroup H_0 determines the covering p completely, up to a suitable notion of equivalence of coverings. This we shall prove in §79. Furthermore, under a (fairly mild) additional "local niceness" condition on B, there exists, for each subgroup H_0 of $\pi_1(B, b_0)$, a covering $p : E \to B$ of B whose corresponding subgroup is H_0. This we shall prove in §82.

Roughly speaking, these results show that one can determine all covering spaces of B merely by examining the collection of all subgroups of $\pi_1(B, b_0)$. This is the classical procedure of algebraic topology; one "solves" a problem of topology by reducing it to a problem of algebra, hopefully one that is more tractable.

Throughout the chapter, we assume the general lifting correspondence theorem, Theorem 54.6.

§79 Equivalence of Covering Spaces

In this section, we show that the subgroup H_0 of $\pi_1(B, b_0)$ determines the covering $p : E \to B$ completely, up to a suitable notion of equivalence of coverings.

Definition. Let $p : E \to B$ and $p' : E' \to B$ be covering maps. They are said to be ***equivalent*** if there exists a homeomorphism $h : E \to E'$ such that $p = p' \circ h$. The homeomorphism h is called an ***equivalence of covering maps*** or an ***equivalence of covering spaces***

$$\begin{tikzcd} E \arrow[rr, "h"] \arrow[dr, "p"'] & & E' \arrow[dl, "p'"] \\ & B & \end{tikzcd}$$

Given two covering maps $p : E \to B$ and $p' : E' \to B$ whose corresponding subgroups H_0 and H_0' are equal, we shall prove that there exists an equivalence $h : E \to E'$. For this purpose, we need to generalize the lifting lemmas of §54.

Lemma 79.1 (The general lifting lemma). *Let $p : E \to B$ be a covering map; let $p(e_0) = b_0$. Let $f : Y \to B$ be a continuous map, with $f(y_0) = b_0$. Suppose Y is path connected and locally path connected. The map f can be lifted to a map $\tilde{f} : Y \to E$ such that $\tilde{f}(y_0) = e_0$ if and only if*

$$f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(E, e_0)).$$

Furthermore, if such a lifting exists, it is unique.

Proof. If the lifting $\tilde{f}$ exists, then

$$f_*(\pi_1(Y, Y_0)) = p_*(\tilde{f}_*(\pi_1(Y, y_0))) \subset p_*(\pi_1(E, e_0)).$$

This proves the "only if" part of the theorem.

Now we prove that if $\tilde{f}$ exists, it is unique. Given $y_1 \in Y$, choose a path α in Y from y_0 to y_1. Take the path $f \circ \alpha$ in B and lift it to a path γ in E beginning at e_0. If a lifting $\tilde{f}$ of f exists, then $\tilde{f}(y_1)$ must equal the end point $\gamma(1)$ of γ, for $\tilde{f} \circ \alpha$ is a lifting of $f \circ \alpha$ that begins at e_0, and path liftings are unique.

Finally, we prove the "if" part of the theorem. The uniqueness part of the proof gives us a clue how to proceed. Given $y_1 \in Y$, choose a path α in Y from y_0 to y_1. Lift the path $f \circ \alpha$ to a path γ in E beginning at e_0, and define $\tilde{f}(y_1) = \gamma(1)$. See Figure 79.1. It is a certain amount of work to show that $\tilde{f}$ is well-defined, independent of the choice of α. Once we prove that, continuity of $\tilde{f}$ is proved easily, as we now show.

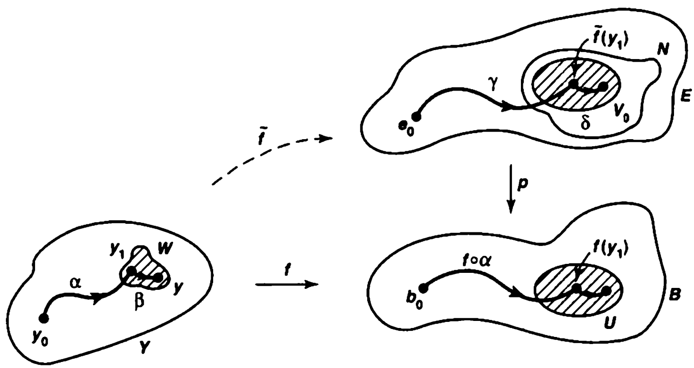

Figure 79.1

To prove continuity of $\tilde{f}$ at the point y_1 of Y, we show that, given a neighborhood N of $\tilde{f}(y_1)$, there is a neighborhood W of y_1 such that $\tilde{f}(W) \subset N$. To begin, choose a path-connected neighborhood U of $f(y_1)$ that is evenly covered by p. Break $p^{-1}(U)$ up into slices, and let V_0 be the slice that contains the point $\tilde{f}(y_1)$. Replacing U by a smaller neighborhood of $f(y_1)$ if necessary, we can assume that $V_0 \subset N$. Let $p_0 : V_0 \to U$ be obtained by restricting p; then p_0 is a homeomorphism. Because f is continuous at y_1 and Y is locally path connected, we can find a path-connected neighborhood W of y_1 such that $f(W) \subset U$. We shall show that $\tilde{f}(W) \subset V_0$; then our result is proved.

Given $y \in W$, choose a path β in W from y_1 to y. Since $\tilde{f}$ is well defined, $\tilde{f}(y)$ can be obtained by taking the path $\alpha * \beta$ from y_0 to y, lifting the path $f \circ (\alpha * \beta)$ to a path in E beginning at e_0, and letting $\tilde{f}(y)$ be the end point of this lifted path. Now γ is a lifting of α that begins at e_0. Since the path $f \circ \beta$ lies in U, the path $\delta = p_0^{-1} \circ f \circ \beta$

is a lifting of it that begins at $\tilde{f}(y_1)$. Then $\gamma * \delta$ is a lifting of $f \circ (\alpha * \beta)$ that begins at e_0; it ends at the point $\delta(1)$ of V_0. Hence $\tilde{f}(W) \subset V_0$, as desired.

Finally, we show $\tilde{f}$ is well defined. Let α and β be two paths in Y from y_0 to y_1. We must show that if we lift $f \circ \alpha$ and $f \circ \beta$ to paths in E beginning at e_0, then these lifted paths end at the same point of E.

First, we lift $f \circ \alpha$ to a path γ in E beginning at e_0; then we lift $f \circ \bar{\beta}$ to a path δ in E beginning at the end point $\gamma(1)$ of γ. Then $\gamma * \delta$ is a lifting of the loop $f \circ (\alpha * \bar{\beta})$. Now by hypothesis,

$$f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(E, e_0)).$$

Hence $[f \circ (\alpha * \bar{\beta})]$ belongs to the image of p_*. Theorem 54.6 now implies that its lift $\gamma * \delta$ is a *loop* in E.

It follows that $\tilde{f}$ is well defined. For $\bar{\delta}$ is a lifting of $f \circ \beta$ that begins at e_0, and γ is a lifting of $f \circ \alpha$ that begins at e_0, and both liftings end at the same point of E. ■

Theorem 79.2. *Let $p : E \to B$ and $p' : E' \to B$ be covering maps; let $p(e_0) = p'(e_0') = b_0$. There is an equivalence $h : E \to E'$ such that $h(e_0) = e_0'$ if and only if the groups*

$$H_0 = p_*(\pi_1(E, e_0)) \qquad \text{and} \qquad H_0' = p_*'(\pi_1(E', e_0'))$$

are equal. If h exists, it is unique.

Proof. We prove the "only if" part of the theorem. Given h, the fact that h is a homeomorphism implies that

$$h_*(\pi_1(E, e_0)) = \pi_1(E', e_0').$$

Since $p' \circ h = p$, we have $H_0 = H_0'$.

Now we prove the "if" part of the theorem; we assume that $H_0 = H_0'$ and show that h exists. We shall apply the preceding lemma (four times!). Consider the maps

$$\begin{tikzcd} & E' \arrow[d, "p'"] \\ E \arrow[r, "p"] & B. \end{tikzcd}$$

Because p' is a covering map and E is path connected and locally path connected, there exists a map $h : E \to E'$ with $h(e_0) = e_0'$ that is a lifting of p (that is, such that $p' \circ h = p$). Reversing the roles of E and E' in this argument, we see there is a map $k : E' \to E$ with $k(e_0') = e_0$ such that $p \circ k = p'$. Now consider the maps

$$\begin{tikzcd} & E \arrow[d, "p"] \\ E \arrow[r, "p"] & B. \end{tikzcd}$$

The map $k \circ h : E \to E$ is a lifting of p (since $p \circ k \circ h = p' \circ h = p$), with $p(e_0) = e_0$. The identity map i_E of E is another such lifting. The uniqueness part of the preceding lemma implies that $k \circ h = i_E$. A similar argument shows that $h \circ k$ equals the identity map of E'. ∎

We seem to have solved our equivalence problem. But there is a somewhat subtle point we have overlooked. We have obtained a necessary and sufficient condition for there to exist an equivalence $h : E \to E'$ that carries the point e_0 to the point e_0'. But we have not yet determined under what conditions there exists an equivalence in general. It is possible that there may be no equivalence carrying e_0 to e_0' but that there *is* an equivalence carrying e_0 to some other point e_1' of $(p')^{-1}(b_0)$. Can we determine whether this is the case merely by examining the subgroups H_0 and H_0'? We consider this problem now.

If H_1 and H_2 are subgroups of a group G, you may recall from algebra that they are said to be ***conjugate*** subgroups if $H_2 = \alpha \cdot H_1 \cdot \alpha^{-1}$ for some element α of G. Said differently, they are conjugate if the isomorphism of G with itself that maps x to $\alpha \cdot x \cdot \alpha^{-1}$ carries the group H_1 onto the group H_2. It is easy to check that conjugacy is an equivalence relation on the collection of subgroups of G. The equivalence class of the subgroup H is called the ***conjugacy class*** of H.

Lemma 79.3. *Let $p : E \to B$ be a covering map. Let e_0 and e_1 be points of $p^{-1}(b_0)$, and let $H_i = p_*(\pi_1(E, e_i))$.*

(a) If γ is a path in E from e_0 to e_1, and α is the loop $p \circ \gamma$ in B, then the equation $[\alpha] * H_1 * [\alpha]^{-1} = H_0$ holds; hence H_0 and H_1 are conjugate.

(b) Conversely, given e_0, and given a subgroup H of $\pi_1(B, b_0)$ conjugate to H_0, there exists a point e_1 of $p^{-1}(b_0)$ such that $H_1 = H$.

Proof. (a) First, we show that $[\alpha] * H_1 * [\alpha]^{-1} \subset H_0$. Given an element $[h]$ of H_1, we have $[h] = p_*([\tilde{h}])$ for some loop $\tilde{h}$ in E based at e_1. Let $\tilde{k}$ be the path $\tilde{k} = (\gamma * \tilde{h}) * \bar{\gamma}$; it is a loop in E based at e_0, and

$$p_*([\tilde{k}]) = [(\alpha * h) * \bar{\alpha}] = [\alpha] * [h] * [\alpha]^{-1},$$

so the latter element belongs to $p_*(\pi_1(E, e_0)) = H_0$, as desired. See Figure 79.2.

Now we show that $[\alpha] * H_1 * [\alpha]^{-1} \supset H_0$. Note that $\bar{\gamma}$ is a path from e_1 to e_0 and $\bar{\alpha}$ equals the loop $p \circ \bar{\gamma}$. By the result just proved, we have

$$[\bar{\alpha}] * H_0 * [\bar{\alpha}]^{-1} \subset H_1,$$

which implies out desired result.

(b) To prove the converse, let e_0 be given and let H be conjugate to H_0. Then $H_0 = [\alpha] * H * [\alpha]^{-1}$ for some loop α in B based at b_0. Let γ be the lifting of α to a path in E beginning at e_0, and let $e_1 = \gamma(1)$. Then (a) implies that $H_0 = [\alpha] * H_1 * [\alpha]^{-1}$. We conclude that $H = H_1$. ∎

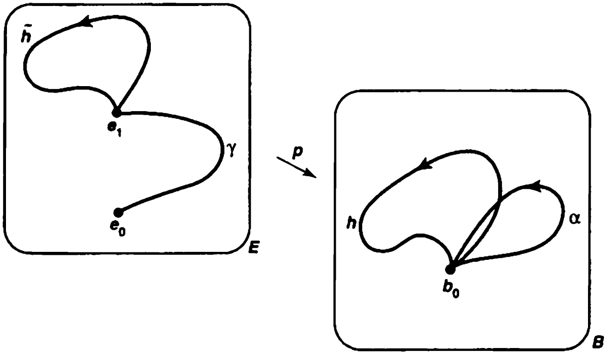

Figure 79.2

Theorem 79.4. *Let $p : E \to B$ and $p' : E' \to B$ be covering maps; let $p(e_0) = p'(e_0') = b_0$. The covering maps p and p' are equivalent if and only if the subgroups*

$$H_0 = p_*(\pi_1(E, e_0)) \quad \text{and} \quad H_0' = p_*'(\pi_1(E', e_0'))$$

of $\pi_1(B, b_0)$ are conjugate.

Proof. If $h : E \to E'$ is an equivalence, let $e_1' = h(e_0)$, and let $H_1' = p_*(\pi_1(E', e_1'))$ Theorem 79.2 implies that $H_0 = H_1'$, while the preceding lemma tells us that H_1' is conjugate to H_0'.

Conversely, if the groups H_0 and H_0' are conjugate, the preceding lemma implies there is a point e_1' of E' such that $H_1' = H_0$. Theorem 79.2 then gives us an equivalence $h : E \to E'$ such that $h(e_0) = e_1'$. ∎

EXAMPLE 1. Consider covering spaces of the circle $B = S^1$. Because $\pi_1(B, b_0)$ is abelian, two subgroups of $\pi_1(B, b_0)$ are conjugate if and only if they are equal. Therefore two coverings of B are equivalent if and only if they correspond to the same subgroup of $\pi_1(B, b_0)$.

Now $\pi_1(B, b_0)$ is isomorphic to the integers $\mathbb{Z}$. What are the subgroups of $\mathbb{Z}$? It is standard theorem of modern algebra that, given a nontrivial subgroup of $\mathbb{Z}$, it must be the group G_n consisting of all multiples of n, for some $n \in \mathbb{Z}_+$.

We have studied one covering space of the circle, the covering $p : \mathbb{R} \to S^1$ It must correspond to the trivial subgroup of $\pi_1(S^1, b_0)$, because $\mathbb{R}$ is simply connected We have also considered the covering $p : S^1 \to S^1$ defined by $p(z) = z^n$, where z is a complex number. In this case, the map p_* carries a generator of $\pi_1(S^1, b_0)$ into n times itself. Therefore, the group $p_*(\pi_1(S^1, b_0))$ corresponds to the subgroup G_n of $\mathbb{Z}$ under the standard isomorphism of $\pi_1(S^1, b_0)$ with $\mathbb{Z}$.

We conclude from the preceding theorem that every path-connected covering space of S^1 is equivalent to one of these coverings.

Exercises

1. Show that if $n > 1$, every continuous map $f : S^n \to S^1$ is nulhomotopic. [*Hint:* Use the lifting lemma.]
2. (a) Show that every continuous map $f : P^2 \to S^1$ is nulhomotopic.
   (b) Find a continuous map of the torus into S^1 that is not nulhomotopic.
3. Let $p : E \to B$ be a covering map; let $p(e_0) = b_0$. Show that $H_0 = p_*(\pi_1(E, e_0))$ is a normal subgroup of $\pi_1(B, b_0)$ if and only if for every pair of points e_1, e_2 of $p^{-1}(b_0)$, there is an equivalence $h : E \to E$ with $h(e_1) = e_2$.
4. Let $T = S^1 \times S^1$, the torus. There is an isomorphism of $\pi_1(T, b_0 \times b_0)$ with $\mathbb{Z} \times \mathbb{Z}$ induced by projections of T onto its two factors.
   (a) Find a covering space of T corresponding to the subgroup of $\mathbb{Z} \times \mathbb{Z}$ generated by the element $m \times 0$, where m is a positive integer.
   (b) Find a covering space of T corresponding to the trivial subgroup of $\mathbb{Z} \times \mathbb{Z}$.
   (c) Find a covering space of T corresponding to the subgroup of $\mathbb{Z} \times \mathbb{Z}$ generated by $m \times 0$ and $0 \times n$, where m and n are positive integers.

*5. Let $T = S^1 \times S^1$ be the torus; let $x_0 = b_0 \times b_0$.
   (a) Prove the following·
   Theorem. *Every isomorphism of $\pi_1(T, x_0)$ with itself is induced by a homeomorphism of T with itself that maps x_0 to x_0.*
   [*Hint:* Let $p : \mathbb{R}^2 \to T$ be the usual covering map. If A is a 2×2 matrix with integer entries, the linear map $T_A : \mathbb{R}^2 \to \mathbb{R}^2$ with matrix A induces a continuous map $f : T \to T$. Furthermore, f is a homeomorphism if A is invertible over the integers.]
   (b) Prove the following:
   Theorem. *If E is a covering space of T, then E is homeomorphic either to $\mathbb{R}^2$, or to $S^1 \times \mathbb{R}$, or to T.*
   [*Hint:* You may use the following result from algebra: If F is a free abelian group of rank 2 and N is a nontrivial subgroup, then there is a basis a_1, a_2 for F such that either (1) ma_1 is a basis for N, for some positive integer m, or (2) ma_1, na_2 is a basis for N, where m and n are positive integers.]

*6. Prove the following:
   Theorem. *Let G be a topological group with multiplication operation $m : G \times G \to G$ and identity element e. Assume $p : \tilde{G} \to G$ is a covering map. Given $\tilde{e}$ with $p(\tilde{e}) = e$, there is a unique multiplication operation on $\tilde{G}$ that makes it into a topological group such that $\tilde{e}$ is the identity element and p is a homomorphism.*
   Proof. Recall that, by our convention, G and $\tilde{G}$ are path connected and locally path connected.
   (a) Let $I : G \to G$ be the map $I(g) = g^{-1}$. Show there exist unique maps $\tilde{m} : \tilde{G} \times \tilde{G} \to \tilde{G}$ and $\tilde{I} : \tilde{G} \to \tilde{G}$ with $\tilde{m}(\tilde{e} \times \tilde{e}) = \tilde{e}$ and $\tilde{I}(\tilde{e}) = \tilde{e}$ such that $p \circ \tilde{m} = m \circ (p \times p)$ and $p \circ \tilde{I} = I \circ p$.
   (b) Show the maps $\tilde{G} \to \tilde{G}$ given by $\tilde{g} \to \tilde{m}(\tilde{e} \times \tilde{g})$ and $\tilde{g} \to \tilde{m}(\tilde{g} \times \tilde{e})$ equal the identity map of $\tilde{G}$. [*Hint:* Use the uniqueness part of Lemma 79.1.]

(c) Show the maps $\tilde{G} \to \tilde{G}$ given by $\tilde{g} \to \tilde{m}(\tilde{g} \times \tilde{I}(\tilde{g}))$ and $\tilde{g} \to \tilde{m}(\tilde{I}(\tilde{g}) \times \tilde{g})$ map $\tilde{G}$ to $\tilde{e}$.

(d) Show the maps $\tilde{G} \times \tilde{G} \times \tilde{G} \to \tilde{G}$ given by

$$\tilde{g} \times \tilde{g}' \times \tilde{g}'' \to \tilde{m}(\tilde{g} \times \tilde{m}(\tilde{g}' \times \tilde{g}''))$$
$$\tilde{g} \times \tilde{g}' \times \tilde{g}'' \to \tilde{m}(\tilde{m}(\tilde{g} \times \tilde{g}') \times \tilde{g}'')$$

are equal.

(e) Complete the proof.

7. Let $p : \tilde{G} \to G$ be a homomorphism of topological groups that is a covering map. Show that if G is abelian, so is $\tilde{G}$.

§80 The Universal Covering Space

Suppose $p : E \to B$ is a covering map, with $p(e_0) = b_0$. If E is simply connected, then E is called a ***universal covering space*** of B. Since $\pi_1(E, e_0)$ is trivial, this covering space corresponds to the trivial subgroup of $\pi_1(B, b_0)$ under the correspondence defined in the preceding section. Theorem 79.4 thus implies that any two universal covering spaces of B are equivalent. For this reason, we often speak of "the" universal covering space of a given space B. Not every space has a universal covering space, as we shall see. For the moment, we shall simply assume that B has a universal covering space and derive some consequences of this assumption.

We prove two preliminary lemmas:

Lemma 80.1. *Let B be path connected and locally path connected. Let $p : E \to B$ be a covering map in the former sense (so that E is not required to be path connected). If E_0 is a path component of E, then the map $p_0 : E_0 \to B$ obtained by restricting p is a covering map.*

Proof. We first show p_0 is surjective. Since the space E is locally homeomorphic to B, it is locally path connected. Therefore E_0 is open in E. It follows that $p(E_0)$ is open in B. We show that $p(E_0)$ is also closed in B, so that $p(E_0) = B$.

Let x be a point of B belonging to the closure of $p(E_0)$. Let U be a path-connected neighborhood of x that is evenly covered by p. Since U contains a point of $p(E_0)$, some slice V_α of $p^{-1}(U)$ must intersect E_0. Since V_α is homeomorphic to U, it is path connected; therefore it must be contained in E_0. Then $p(V_\alpha) = U$ is contained in $p(E_0)$, so that in particular, $x \in p(E_0)$.

Now we show $p_0 : E_0 \to B$ is a covering map. Given $x \in B$, choose a neighborhood U of x as before. If V_α is a slice of $p^{-1}(U)$, then V_α is path connected; if it intersects E_0, it lies in E_0. Therefore, $p_0^{-1}(U)$ equals the union of those slices V_α of $p^{-1}(U)$ that intersect E_0; each of these is open in E_0 and is mapped homeomorphically by p_0 onto U. Thus U is evenly covered by p_0. ∎

Lemma 80.2. *Let p, q, and r be continuous maps with $p = r \circ q$, as in the following diagram:*

$$\begin{tikzcd} X \arrow[d, "p"'] \arrow[dr, "q"] & \\ Z & Y \arrow[l, "r"] \end{tikzcd}$$

(a) If p and r are covering maps, so is q.

**(b) If p and q are covering maps, so is r.*

Proof. By our convention, X, Y, and Z are path connected and locally path connected. Let $x_0 \in X$; set $y_0 = q(x_0)$ and $z_0 = p(x_0)$.

(a) Assume that p and r are covering maps. We show first that q is surjective. Given $y \in Y$, choose a path $\tilde{\alpha}$ in Y from y_0 to y. Then $\alpha = r \circ \tilde{\alpha}$ is a path in Z beginning at z_0; let $\tilde{\tilde{\alpha}}$ be a lifting of α to a path in X beginning at x_0. Then $q \circ \tilde{\tilde{\alpha}}$ is a lifting of α to Y that begins at y_0. By uniqueness of path liftings, $\tilde{\alpha} = q \circ \tilde{\tilde{\alpha}}$. Then q maps the end point of $\tilde{\tilde{\alpha}}$ to the end point y of $\tilde{\alpha}$. Thus q is surjective.

Given $y \in Y$, we find a neighborhood of y that is evenly covered by q. Let $z = r(y)$. Since p and r are covering maps, we can find a path-connected neighborhood U of z that is evenly covered by both p and r. Let V be the slice of $r^{-1}(U)$ that contains the point y; we show V is evenly covered by q. Let $\{U_\alpha\}$ be the collection of slices of $p^{-1}(U)$. Now q maps each set U_α into the set $r^{-1}(U)$; because U_α is connected, it must be mapped by q into a single one of the slices of $r^{-1}(U)$. Therefore, $q^{-1}(V)$ equals the union of those slices U_α that are mapped by q into V. It is easy to see that each such U_α is mapped homeomorphically *onto* V by q. For let p_0, q_0, r_0 be the maps obtained by restricting p, q, and r, respectively, as indicated in the following diagram:

$$\begin{tikzcd} U_\alpha \arrow[d, "p_0"'] \arrow[dr, "q_0"] & \\ U & V \arrow[l, "r_0"] \end{tikzcd}$$

Because p_0 and r_0 are homeomorphisms, so is $q_0 = r_0^{-1} \circ p_0$.

*(b) We shall use this result only in the exercises. Assume that p and q are covering maps. Because $p = r \circ q$ and p is surjective, r is also surjective.

Given $z \in Z$, let U be a path-connected neighborhood of z that is evenly covered by p. We show that U is also evenly covered by r. Let $\{V_\beta\}$ be the collection of path components of $r^{-1}(U)$; these sets are disjoint and open in Y. We show that for each β, the map r carries V_β homeomorphically onto U.

Let $\{U_\alpha\}$ be the collection of slices of $p^{-1}(U)$; they are disjoint, open, and path connected, so they are the path components of $p^{-1}(U)$. Now q maps each U_α into the set $r^{-1}(U)$; because U_α is connected, it must be mapped by q into one of the sets V_β. Therefore $q^{-1}(V_\beta)$ equals the union of a subcollection of the collection $\{U_\alpha\}$. Theorem 53.2 and Lemma 80.1 together imply that if U_{α_0} is any one of the path components of $q^{-1}(V_\beta)$ then the map $q_0 : U_{\alpha_0} \to V_\beta$ obtained by restricting q is a covering map.

In particular, q_0 is surjective. Hence q_0 is a homeomorphism, being continuous, open, and injective as well. Consider the maps

$$\begin{tikzcd} U_{\alpha_0} \arrow[d, "p_0"'] \arrow[dr, "q_0"] & \\ U & V_\beta \arrow[l, "r_0"] \end{tikzcd}$$

obtained by restricting p, q, and r. Because p_0 and q_0 are homeomorphisms, so is r_0. ∎

Theorem 80.3. *Let $p : E \to B$ be a covering map, with E simply connected. Given any covering map $r : Y \to B$, there is a covering map $q : E \to Y$ such that $r \circ q = p$.*

$$\begin{tikzcd} E \arrow[d, "p"'] \arrow[dr, "q"] & \\ B & Y \arrow[l, "r"] \end{tikzcd}$$

This theorem shows why E is called a *universal* covering space of B; it covers every other covering space of B.

Proof. Let $b_0 \in B$; choose e_0 and y_0 so that $p(e_0) = b_0$ and $r(y_0) = b_0$. We apply Lemma 79.1 to construct q. The map r is a covering map, and the condition

$$p_*(\pi_1(E, e_0)) \subset r_*(\pi_1(Y, y_0))$$

is satisfied trivially because E is simply connected. Therefore, there is a map $q : E \to Y$ such that $r \circ q = p$ and $q(e_0) = y_0$. It follows from the preceding lemma that q is a covering map. ∎

Now we give an example of a space that has no universal covering space. We need the following lemma.

Lemma 80.4. *Let $p : E \to B$ be a covering map; let $p(e_0) = b_0$. If E is simply connected, then b_0 has a neighborhood U such that inclusion $i : U \to B$ induces the trivial homomorphism*

$$i_* : \pi_1(U, b_0) \longrightarrow \pi_1(B, b_0).$$

Proof. Let U be a neighborhood of b_0 that is evenly covered by p; break $p^{-1}(U)$ up into slices; let U_α be the slice containing e_0. Let f be a loop in U based at b_0. Because p defines a homeomorphism of U_α with U, the loop f lifts to a *loop* $\tilde{f}$ in U_α based at e_0. Since E is simply connected, there is a path homotopy $\tilde{F}$ in E between $\tilde{f}$ and a constant loop. Then $p \circ \tilde{F}$ is a path homotopy in B between f and a constant loop. ∎

EXAMPLE 1. Let X be our familiar "infinite earring" in the plane, if C_n is the circle of radius $1/n$ in the plane with center at the point $(1/n, 0)$, then X is the union of the circles C_n. Let b_0 be the origin; we show that if U is any neighborhood of b_0 in X, then the homomorphism of fundamental groups induced by inclusion $i : U \to X$ is not trivial.

Given n, there is a retraction $r : X \to C_n$ obtained by letting r map each circle C_i for $i \neq n$ to the point b_0. Choose n large enough that C_n lies in U. Then in the following diagram of homomorphisms induced by inclusion, j_* is injective, hence i_* cannot be trivial.

$$\begin{tikzcd} \pi_1(C_n, b_0) \arrow[rr, "j_*"] \arrow[dr, "k_*"'] & & \pi_1(X, b_0) \\ & \pi_1(U, b_0) \arrow[ur, "i_*"'] & \end{tikzcd}$$

It follows that even though X is path connected and locally path connected, it has no universal covering space.

Exercise

1. Let $q : X \to Y$ and $r : Y \to Z$ be maps; let $p = r \circ q$.
   (a) Let q and r be covering maps. Show that if Z has a universal covering space, then p is a covering map. Compare Exercise 4 of §53.
   *(b) Give an example where q and r are covering maps but p is not.

*§81 Covering Transformations

Given a covering map $p : E \to B$, it is of some interest to consider the set of all equivalences of this covering space with *itself*. Such an equivalence is called a ***covering transformation***. Composites and inverses of covering transformations are covering transformations, so this set forms a group; it is called the ***group of covering transformations*** and denoted $\mathcal{C}(E, p, B)$.

Throughout this section, we shall assume that $p : E \to B$ is a covering map with $p(e_0) = b_0$; and we shall let $H_0 = p_*(\pi_1(E, e_0))$. We shall show that the group $\mathcal{C}(E, p, B)$ is completely determined by the group $\pi_1(B, b_0)$ and the subgroup H_0. Specifically, we shall show that if $N(H_0)$ is the largest subgroup of $\pi_1(B, b_0)$ of which H_0 is a normal subgroup, then $\mathcal{C}(E, p, B)$ is isomorphic to $N(H_0)/H_0$.

We define $N(H_0)$ formally as follows:

Definition. If H is a subgroup of the group G, then the ***normalizer*** of H in G is the subset of G defined by the equation

$$N(H) = \{g \mid gHg^{-1} = H\}.$$

It is easy to see that $N(H)$ is a subgroup of G. It follows from the definition that it contains H as a normal subgroup and is the largest such subgroup of G.

The correspondence between the groups $N(H_0)/H_0$ and $\mathcal{C}(E, p, B)$ is established by using the lifting correspondence of §54 and the results about the existence of equivalences proved in §79. We make the following definition:

Definition. Given $p : E \to B$ with $p(e_0) = b_0$, let F be the set $F = p^{-1}(e_0)$. Let

$$\Phi : \pi_1(B, b_0)/H_0 \to F$$

be the lifting correspondence of Theorem 54.6; it is a bijection. Define also a correspondence

$$\Psi : \mathcal{C}(E, p, B) \to F$$

by setting $\Psi(h) = h(e_0)$ for each covering transformation $h : E \to E$. Since h is uniquely determined once its value at e_0 is known, the correspondence Ψ is injective.

Lemma 81.1. *The image of the map Ψ equals the image under Φ of the subgroup $N(H_0)/H_0$ of $\pi_1(B, b_0)/H_0$.*

Proof. Recall that the lifting correspondence $\phi : \pi_1(B, b_0) \to F$ is defined as follows: Given a loop α in B at b_0, let γ be its lift to E beginning at e_0; let $e_1 = \gamma(1)$; and define ϕ by setting $\phi([\alpha]) = e_1$. To prove the lemma, we need to show that there is a covering transformation $h : E \to E$ with $h(e_0) = e_1$ if and only if $[\alpha] \in N(H_0)$.

This is easy. Lemma 79.1 tells us that h exists if and only if $H_0 = H_1$, where $H_1 = p_*(\pi_1(E, e_1))$. And Lemma 79.3 tells us that $[\alpha] * H_1 * [\alpha]^{-1} = H_0$. Hence h exists if and only if $[\alpha] * H_0 * [\alpha]^{-1} = H_0$, which is simply the statement that $[\alpha] \in N(H_0)$. ■

Theorem 81.2. *The bijection*

$$\Phi^{-1} \circ \Psi : \mathcal{C}(E, p, B) \to N(H_0)/H_0$$

is an isomorphism of groups.

Proof. We need only show that $\Phi^{-1} \circ \Psi$ is a homomorphism. Let $h, k : E \to E$ be covering transformations. Let $h(e_0) = e_1$ and $k(e_0) = e_2$; then

$$\Psi(h) = e_1 \quad \text{and} \quad \Psi(k) = e_2,$$

by definition. Choose paths γ and δ in E from e_0 to e_1 and e_2, respectively. If $\alpha = p \circ \gamma$ and $\beta = p \circ \delta$, then

$$\Phi([\alpha]H_0) = e_1 \quad \text{and} \quad \Phi([\beta]H_0) = e_2,$$

by definition. Let $e_3 = h(k(e_0))$; then $\Psi(h \circ k) = e_3$. We show that

$$\Phi([\alpha * \beta]H_0) = e_3,$$

and the proof is complete.

Since δ is a path from e_0 to e_2, the path $h \circ \delta$ is a path from $h(e_0) = e_1$ to $h(e_2) = h(k(e_0)) = e_3$. See Figure 81.1. Then the product $\gamma * (h \circ \delta)$ is defined and is a path from e_0 to e_3. It is a lifting of $\alpha * \beta$, since $p \circ \gamma = \alpha$ and $p \circ h \circ \delta = p \circ \delta = \beta$. Therefore $\Phi([\alpha * \beta]H_0) = e_3$, as desired. ∎

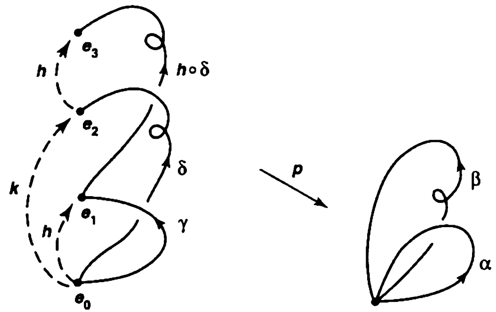

Figure 81.1

Corollary 81.3. *The group H_0 is a normal subgroup of $\pi_1(B, b_0)$ if and only if for every pair of points e_1 and e_2 of $p^{-1}(b_0)$, there is a covering transformation $h : E \to E$ with $h(e_1) = e_2$. In this case, there is an isomorphism*

$$\Phi^{-1} \circ \Psi : \mathcal{C}(E, p, B) \to \pi_1(B, b_0)/H_0.$$

Corollary 81.4. *Let $p : E \to B$ be a covering map. If E is simply connected, then*

$$\mathcal{C}(E, p, B) \cong \pi_1(B, b_0).$$

If H_0 is a normal subgroup of $\pi_1(B, b_0)$, then $p : E \to B$ is called a ***regular covering map*** (Here is another example of the overuse of familiar terms. The words "normal" and "regular" have already been used to mean quite different things!)

EXAMPLE 1. Because the fundamental group of the circle is abelian, every covering of S^1 is regular. If $p : \mathbb{R} \to S^1$ is the standard covering map, for instance, the covering transformations are the homeomorphisms $x \to x + n$. The group of such transformations is isomorphic to $\mathbb{Z}$.

EXAMPLE 2 For an example at the other extreme, consider the covering space of the figure eight indicated in Figure 81.2. (We considered this covering earlier, in §60. The x-axis is wrapped around the circle A and the y-axis is wrapped around B. The circles A_i and B_i are mapped homeomorphically onto A and B, respectively.) In this case, we show that the group $\mathcal{C}(E, p, B)$ is trivial.

In general, if $h : E \to E$ is a covering transformation, then any loop in the base space that lifts to a loop in E at e_0 also lifts to a loop when the lift begins at $h(e_0)$. In the present case, a loop that generates the fundamental group of A lifts to a non-loop when the lift is based at e_0 and lifts to a loop when it is based at any other point of $p^{-1}(b_0)$ lying on the y-axis. Similarly, a loop that generates the fundamental group of B lifts to a non-loop beginning at e_0 and to a loop beginning at any other point of $p^{-1}(b_0)$ lying on the x-axis. It follows that $h(e_0) = e_0$, so that h is the identity map.

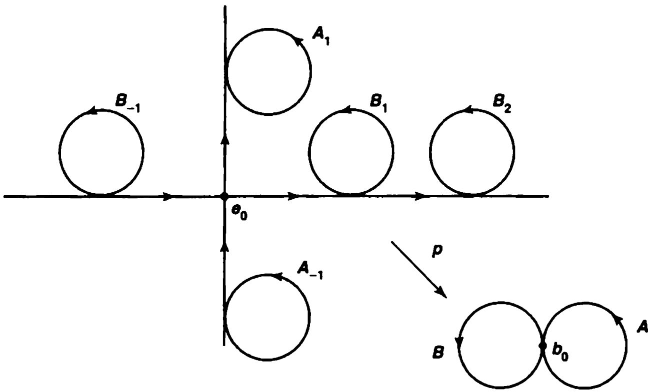

Figure 81.2

There is a method for constructing covering spaces that automatically leads to a covering that is regular; and in fact every regular covering space can be constructed by this method. It involves the *action* of a group G on a space X.

Definition. Let X be a space, and let G be a subgroup of the group of homeomorphisms of X with itself. The ***orbit space*** X/G is defined to be the quotient space obtained from X by means of the equivalence relation $x \sim g(x)$ for all $x \in X$ and all $g \in G$. The equivalence class of x is called the ***orbit*** of x.

Definition. If G is a group of homeomorphisms of X, the action of G on X is said to be ***properly discontinuous*** if for every $x \in X$ there is a neighborhood U of x such that $g(U)$ is disjoint from U whenever $g \neq e$. (Here e is the identity element of G.) It follows that $g_0(U)$ and $g_1(U)$ are disjoint whenever $g_0 \neq g_1$, for otherwise U and $g_0^{-1}g_1(U)$ would not be disjoint.

Theorem 81.5. *Let X be path connected and locally path connected; let G be a group of homeomorphisms of X. The quotient map $\pi : X \to X/G$ is a covering map if and only if the action of G is properly discontinuous. In this case, the covering map π is regular and G is its group of covering transformations.*

Proof. We show π is an open map. If U is open in X, then $\pi^{-1}\pi(U)$ is the union of the open sets $g(U)$ of X, for $g \in G$. Hence $\pi^{-1}\pi(U)$ is open in X, so that $\pi(U)$ is open in X/G by definition. Thus π is open.

Step 1. We suppose that the action of G is properly discontinuous and show that π is a covering map. Given $x \in X$, let U be a neighborhood of x such that $g_0(U)$ and $g_1(U)$ are disjoint whenever $g_0 \neq g_1$. Then $\pi(U)$ is evenly covered by π. Indeed, $\pi^{-1}\pi(U)$ equals the union of the disjoint open sets $g(U)$, for $g \in G$, each of which contains at most one point of each orbit. Therefore, the map $g(U) \to \pi(U)$ obtained by restricting π is bijective; being continuous and open, it is a homeomorphism. The sets $g(U)$, for $g \in G$, thus form a partition of $\pi^{-1}\pi(U)$ into slices.

Step 2. We suppose now that π is a covering map and show that the action of G is properly discontinuous. Given $x \in X$, let V be a neighborhood of $\pi(x)$ that is evenly covered by π. Partition $\pi^{-1}(V)$ into slices; let U_α be the slice containing x. Given $g \in G$ with $g \neq e$, the set $g(U_\alpha)$ must be disjoint from U_α, for otherwise, two points of U_α would belong to the same orbit and the restriction of π to U_α would not be injective. It follows that the action of G is properly discontinuous.

Step 3. We show that if π is a covering map, then G is its group of covering transformations and π is regular. Certainly any $g \in G$ is a covering transformation, for $\pi \circ g = \pi$ because the orbit of $g(x)$ equals the orbit of x. On the other hand, let h be a covering transformation with $h(x_1) = x_2$, say. Because $\pi \circ h = \pi$, the points x_1 and x_2 map to the same point under π; therefore there is an element $g \in G$ such that $g(x_1) = x_2$. The uniqueness part of Theorem 79.2 then implies that $h = g$.

It follows that π is regular. Indeed, for any two points x_1 and x_2 lying in the same orbit, there is an element $g \in G$ such that $g(x_1) = x_2$. Then Corollary 81.3 applies. ∎

Theorem 81.6. *If $p : X \to B$ is a regular covering map and G is its group of covering transformations, then there is a homeomorphism $k : X/G \to B$ such that $p = k \circ \pi$, where $\pi : X \to X/G$ is the projection.*

$$\begin{tikzcd} X \arrow[r, phantom, "="] \arrow[d, "\pi"] & X \arrow[d, "p"] \\ X/G \arrow[r, "k"] & B \end{tikzcd}$$

Proof. If g is a covering transformation, then $p(g(x)) = p(x)$ by definition. Hence p is constant on each orbit, so it induces a continuous map k of the quotient space X/G into B. On the other hand, p is a quotient map because it is continuous, surjective, and open. Because p is regular, any two points of $p^{-1}(b)$ belong to the same orbit under the action of G. Therefore, π induces a continuous map $B \to X/G$ that is an inverse for k. ∎

EXAMPLE 3. Let X be the cylinder $S^1 \times I$; let $h : X \to X$ be the homeomorphism $h(x, t) = (-x, t)$; and let $k : X \to X$ be the homeomorphism $k(x, t) = (-x, 1 - t)$. The groups $G_1 = \{e, h\}$ and $G_2 = \{e, k\}$ are isomorphic to the integers modulo 2; both aci properly discontinuously on X. But X/G_1 is homeomorphic to X, while X/G_2 is homeomorphic to the Möbius band, as you can check. See Figure 81.3.

Figure 81.3

Exercises

1. (a) Find a group G of homeomorphisms of the torus T having order 2 such that T/G is homeomorphic to the torus.
   (b) Find a group G of homeomorphisms of T having order 2 that T/G is homeomorphic to the Klein bottle.
2. Let $X = A \vee B$ be the wedge of two circles.
   (a) Let E be the space pictured in Figure 81.4; let $p : E \to X$ wrap each arc A_1 and A_2 around A and map B_1 and B_2 homeomorphically onto B. Show p is a regular covering map.
   (b) Determine the group of covering transformations of the covering of X indicated in Figure 81.5 Is this covering regular?

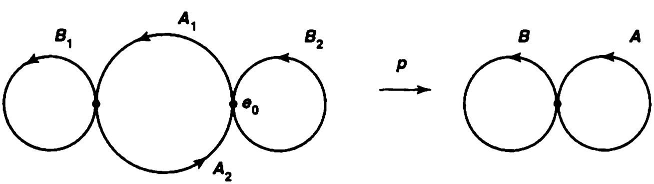

Figure 81.4

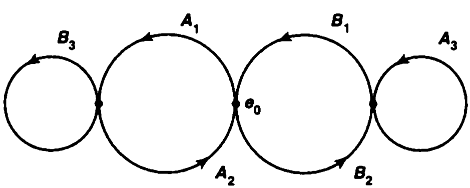

Figure 81.5

(c) Repeat (b) for the covering pictured in Figure 81.6.
(d) Repeat (b) for the covering pictured in Figure 81.7.

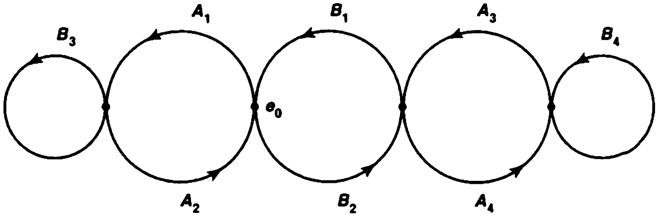

Figure 81.6

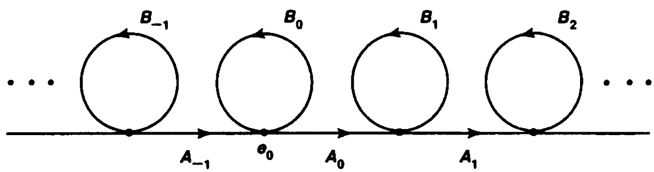

Figure 81.7

3. Let $p : X \to B$ be a covering map (not necessarily regular); let G be its group of covering transformations.
   (a) Show that the action of G on X is properly discontinuous.
   (b) Let $\pi : X \to X/G$ be the projection map. Show there is a covering map $k : X/G \to B$ such that $k \circ \pi = p$.

$$\begin{tikzcd} X \arrow[dr, "\pi"] \arrow[dd, "p"'] & \\ & X/G \arrow[dl, "k"] \\ B & \end{tikzcd}$$

4. Let G be a group of homeomorphisms of X. The action of G on X is said to be ***fixed-point free*** if no element of G other than the identity e has a fixed point. Show that if X is Hausdorff, and if G is a finite group of homeomorphisms of X whose action is fixed-point free, then the action of G is properly discontinuous.

5. Consider S^3 as the space of all pairs of complex numbers (z_1, z_2) satisfying the equation $|z_1|^2 + |z_2|^2 = 1$. Given relatively prime positive integers n and k, define $h : S^3 \to S^3$ by the equation

$$h(z_1, z_2) = (z_1 e^{2\pi i/n}, z_2 e^{2\pi i k/n}).$$

(a) Show that h generates a subgroup G of the homeomorphism group of S^3 that is cyclic of order n, and that only the identity element of G has a fixed point. The orbit space S^3/G is called the ***lens space*** $L(n, k)$.
(b) Show that if $L(n, k)$ and $L(n', k')$ are homeomorphic, then $n = n'$. [It is a theorem that $L(n, k)$ and $L(n', k')$ are homeomorphic if and only if $n = n'$ and either $k \equiv k' \pmod{n}$ or $kk' \equiv 1 \pmod{n}$. The proof is decidedly nontrivial.]
(c) Show that $L(n, k)$ is a compact 3-manifold.

6. Prove the following:
Theorem. *Let X be a locally compact Hausdorff space; let G be a group of homeomorphisms of X such that the action of G is fixed-point free. Suppose that for each compact subspace C of X, there are only finitely many elements g of G such that the intersection $C \cap g(C)$ is nonempty. Then the action of G is properly discontinuous, and X/G is locally compact Hausdorff.*
Proof.
(a) For each compact subspace C of X, show that the union of the sets $g(C)$, for $g \in G$, is closed in X. [*Hint:* If U is a neighborhood of x with $\bar{U}$ compact, then $\bar{U} \cup C$ intersects $g(\bar{U} \cup C)$ for only finitely many g.]
(b) Show X/G is Hausdorff.
(c) Show the action of G is properly discontinuous.
(d) Show X/G is locally compact.

§82 Existence of Covering Spaces

We have shown that corresponding to each covering map $p : E \to B$ is a conjugacy class of subgroups of $\pi_1(B, b_0)$, and that two such covering maps are equivalent if and only if they correspond to the same such class. Thus, we have an injective correspondence from equivalence classes of coverings of B to conjugacy classes of subgroups of $\pi_1(B, b_0)$. Now we ask the question whether this correspondence is surjective, that is, whether for every conjugacy class of subgroups of $\pi_1(B, b_0)$, there exists a covering of B that corresponds to this class.

The answer to this question is "no," in general. In §80, we gave an example of a path-connected, locally path-connected space B that had no simply connected covering space, that is, that had no covering space corresponding to the class of the trivial subgroup. This example relied on Lemma 80.4, which gave a condition that any space having a simply connected covering space must satisfy. We now introduce this condition formally.

Definition. A space B is said to be ***semilocally simply connected*** if for each $b \in B$, there is a neighborhood U of b such that the homomorphism

$$i_* : \pi_1(U, b) \to \pi_1(B, b)$$

induced by inclusion is trivial.

Note that if U satisfies this condition, then so does any smaller neighborhood of b, so that b has "arbitrarily small" neighborhoods satisfying this condition. Note also that this condition is weaker than true local simple connectedness, which would require that within each neighborhood of b there should exist a neighborhood U of b that is itself simply connected.

Semilocal simple connectedness of B is both necessary and sufficient for there to exist, for every conjugacy class of subgroups of $\pi_1(B, b_0)$, a corresponding covering space of B. Necessity was proved in Lemma 80.4; sufficiency is proved in this section.

Theorem 82.1. *Let B be path connected, locally path connected, and semilocally simply connected. Let $b_0 \in B$. Given a subgroup H of $\pi_1(B, b_0)$, there exists a covering map $p : E \to B$ and a point $e_0 \in p^{-1}(b_0)$ such that*

$$p_*(\pi_1(E, e_0)) = H.$$

Proof. Step 1. Construction of E. The procedure for constructing E is reminiscent of the procedure used in complex analysis for constructing Riemann surfaces. Let $\mathcal{P}$ denote the set of all paths in B beginning at b_0. Define an equivalence relation on $\mathcal{P}$ by setting $\alpha \sim \beta$ if α and β end at the same point of B and

$$[\alpha * \bar{\beta}] \in H.$$

This relation is easily seen to be an equivalence relation. We will denote the equivalence class of the path α by $\alpha^\#$.

Let E denote the collection of equivalence classes, and define $p : E \to B$ by the equation

$$p(\alpha^\#) = \alpha(1)$$

Since B is path connected, p is surjective. We shall topologize E so that p is a covering map.

We first note two facts:

(1) If $[\alpha] = [\beta]$, then $\alpha^\# = \beta^\#$.

(2) If $\alpha^\# = \beta^\#$, then $(\alpha * \delta)^\# = (\beta * \delta)^\#$ for any path δ in B beginning at $\alpha(1)$.

The first follows by noting that if $[\alpha] = [\beta]$, then $[\alpha * \bar{\beta}]$ is the identity element, which belongs to H. The second follows by noting that $\alpha * \delta$ and $\beta * \delta$ end at the same point of B, and

$$[(\alpha * \delta) * \overline{(\beta * \delta)}] = [(\alpha * \delta) * (\bar{\delta} * \bar{\beta})] = [\alpha * \bar{\beta}],$$

which belongs to H by hypothesis.

Step 2. Topologizing E. One way to topologize E is to give $\mathcal{P}$ the compact-open topology (see Chapter 7) and E the corresponding quotient topology. But we can topologize E directly as follows:

Let α be any element of $\mathcal{P}$, and let U be any path-connected neighborhood of $\alpha(1)$. Define

$$B(U, \alpha) = \{(\alpha * \delta)^{\#} \mid \delta \text{ is a path in } U \text{ beginning at } \alpha(1)\}.$$

Note that $\alpha^{\#}$ is an element of $B(U, \alpha)$, since if $b = \alpha(1)$, then $\alpha^{\#} = (\alpha * e_b)^{\#}$, this element belongs to $B(U, \alpha)$ by definition. We assert that the sets $B(U, \alpha)$ form a basis for a topology on E.

First, we show that if $\beta^{\#} \in B(U, \alpha)$, then $\alpha^{\#} \in B(U, \beta)$ and $B(U, \alpha) = B(U, \beta)$. If $\beta^{\#} \in B(U, \alpha)$, then $\beta^{\#} = (\alpha * \delta)^{\#}$ for some path δ in U. Then

$$\begin{aligned}(\beta * \bar{\delta})^{\#} &= ((\alpha * \delta) * \bar{\delta})^{\#} && \text{by (2)}\\ &= \alpha^{\#} && \text{by (1),}\end{aligned}$$

so that $\alpha^{\#} \in B(U, \beta)$ by definition. See Figure 82.1. We show first that $B(U, \beta) \subset B(U, \alpha)$. Note that the general element of $B(U, \beta)$ is of the form $(\beta * \gamma)^{\#}$, where γ is a path in U. Then note that

$$\begin{aligned}(\beta * \gamma)^{\#} &= ((\alpha * \delta) * \gamma)^{\#}\\ &= (\alpha * (\delta * \gamma))^{\#},\end{aligned}$$

which belongs to $B(U, \alpha)$ by definition. Symmetry gives the inclusion $B(U, \alpha) \subset B(U, \beta)$ as well.

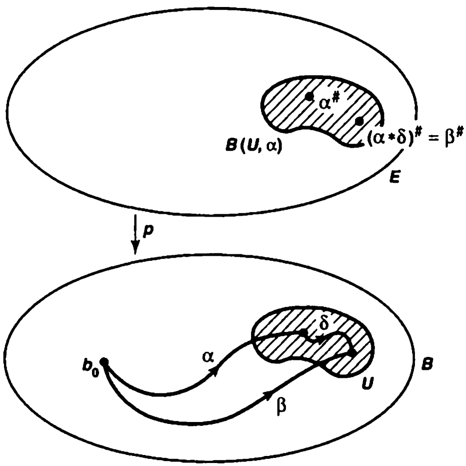

Figure 82.1

Now we show the sets $B(U, \alpha)$ form a basis. If $\beta^{\#}$ belongs to the intersection $B(U_1, \alpha_1) \cap B(U_2, \alpha_2)$, we need merely choose a path-connected neighborhood V

of $\beta(1)$ contained in $U_1 \cap U_2$. The inclusion

$$B(V, \beta) \subset B(U_1, \beta) \cap B(U_2, \beta)$$

follows from the definition of these sets, and the right side of the equation equals $B(U_1, \alpha_1) \cap B(U_2, a_2)$ by the result just proved.

Step 3. The map p is continuous and open. It is easy to see that p is open, for the image of the basis element $B(U, \alpha)$ is the open subset U of B: Given $x \in U$, we choose a path δ in U from $\alpha(1)$ to x; then $(\alpha * \delta)^\#$ is in $B(U, \alpha)$ and $p((\alpha * \delta)^\#) = x$.

To show that p is continuous, let us take an element $\alpha^\#$ of E and a neighborhood W of $p(\alpha^\#)$. Choose a path-connected neighborhood U of the point $p(\alpha^\#) = \alpha(1)$ lying in W. Then $B(U, \alpha)$ is a neighborhood of $\alpha^\#$ that p maps into W. Thus p is continuous at $\alpha^\#$.

Step 4. Every point of B has a neighborhood that is evenly covered by p. Given $b_1 \in B$, choose U to be a path-connected neighborhood of b_1 that satisfies the further condition that the homomorphism $\pi_1(U, b_1) \to \pi_1(B, b_1)$ induced by inclusion is trivial. We assert that U is evenly covered by p.

First, we show that $p^{-1}(U)$ equals the union of the sets $B(U, \alpha)$, as α ranges over all paths in B from b_0 to b_1. Since p maps each set $B(U, \alpha)$ onto U, it is clear that $p^{-1}(U)$ contains this union. On the other hand, if $\beta^\#$ belongs to $p^{-1}(U)$, then $\beta(1) \in U$. Choose a path δ in U from b_1 to $\beta(1)$ and let α be the path $\beta * \bar{\delta}$ from b_0 to b_1. Then $[\beta] = [\alpha * \delta]$, so that $\beta^\# = (\alpha * \delta)^\#$, which belongs to $B(U, \alpha)$. Thus $p^{-1}(U)$ is contained in the union of the sets $B(U, \alpha)$.

Second, note that distinct sets $B(U, \alpha)$ are disjoint. For if $\beta^\#$ belongs to $B(U, \alpha_1) \cap B(U, \alpha_2)$, then $B(U, \alpha_1) = B(U, \beta) = B(U, \alpha_2)$, by Step 2.

Third, we show that p defines a bijective map of $B(U, \alpha)$ with U. It follows that $p|B(U, \alpha)$ is a homeomorphism, being bijective and continuous and open. We already know that p maps $B(U, \alpha)$ *onto* U. To prove injectivity, suppose that

$$p((\alpha * \delta_1)^\#) = p((\alpha * \delta_2)^\#),$$

where δ_1 and δ_2 are paths in U. Then $\delta_1(1) = \delta_2(1)$. Because the homomorphism $\pi_1(U, b_1) \to \pi_1(B, b_1)$ induced by inclusion is trivial, $\delta_1 * \bar{\delta}_2$ is path homotopic in B to the constant loop. Then $[\alpha * \delta_1] = [\alpha * \delta_2]$, so that $(\alpha * \delta_1)^\# = (\alpha * \delta_2)^\#$, as desired.

It follows that $p : E \to B$ is a covering map in the sense used in earlier chapters. To show it is a covering map in the sense used in this chapter, we must show E is path connected. This we shall do shortly.

Step 5. Lifting a path in B. Let e_0 denote the equivalence class of the constant path at b_0; then $p(e_0) = b_0$ by definition. Given a path α in B beginning at b_0, we calculate its lift to a path in E beginning at e_0 and show that this lift ends at $\alpha^\#$.

To begin, given $c \in [0, 1]$, let $\alpha_c : I \to B$ denote the path defined by the equation

$$\alpha_c(t) = \alpha(tc) \quad \text{for} \quad 0 \le t \le 1.$$

Then α_c is the "portion" of α that runs from $\alpha(0)$ to $\alpha(c)$. In particular, α_0 is the constant path at b_0, and α_1 is the path α itself. We define $\tilde{\alpha} : I \to E$ by the equation

$$\tilde{\alpha}(c) = (\alpha_c)^\#$$

and show that $\tilde{\alpha}$ is continuous. Then $\tilde{\alpha}$ is a lift of α, since $p(\tilde{\alpha}(c)) = \alpha_c(1) = \alpha(c)$; furthermore, $\tilde{\alpha}$ begins at $(\alpha_0)^\# = e_0$ and ends at $(\alpha_1)^\# = \alpha^\#$.

To verify continuity, we introduce the following notation. Given $0 \leq c < d \leq 1$, let $\delta_{c,d}$ denote the path that equals the positive linear map of I onto $[c, d]$ followed by α. Note that the paths α_d and $\alpha_c * \delta_{c,d}$ are path homotopic because one is just a reparametrization of the other. See Figure 82.2.

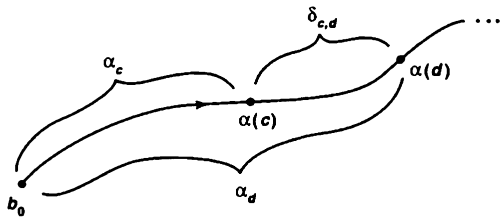

Figure 82.2

We now verify continuity of $\tilde{\alpha}$ at the point c of $[0, 1]$. Let W be a basis element in E about the point $\tilde{\alpha}(c)$. Then W equals $B(U, \alpha_c)$ for some path-connected neighborhood U of $\alpha(c)$. Choose $\epsilon > 0$ so that for $|c - t| < \epsilon$, the point $\alpha(t)$ lies in U. We show that if d is a point of $[0, 1]$ with $|c - d| < \epsilon$, then $\tilde{\alpha}(d) \in W$; this proves continuity of $\tilde{\alpha}$ at c.

So suppose $|c - d| < \epsilon$. Take first the case where $d > c$. Set $\delta = \delta_{c,d}$; then since $[\alpha_d] = [\alpha_c * \delta]$, we have

$$\tilde{\alpha}(d) = (\alpha_d)^\# = (\alpha_c * \delta)^\#.$$

Since δ lies in U, we have $\tilde{\alpha}(d) \in B(U, \alpha_c)$, as desired. If $d < c$, set $\delta = \delta_{d,c}$ and proceed similarly.

Step 6. The map $p : E \to B$ is a covering map. We need only verify that E is path connected, and this is easy. For if $\alpha^\#$ is any point of E, then the lift $\tilde{\alpha}$ of the path α is a path in E from e_0 to $\alpha^\#$.

Step 7. Finally, $H = p_*(\pi_1(E, e_0))$. Let α be a loop in B at b_0. Let $\tilde{\alpha}$ be its lift to E beginning at e_0. Theorem 54.6 tells us that $[\alpha] \in p_*(\pi_1(E, e_0))$ if and only if $\tilde{\alpha}$ is a *loop* in E. Now the final point of $\tilde{\alpha}$ is the point $\alpha^\#$, and $\alpha^\# = e_0$ if and only if α is equivalent to the constant path at b_0, i.e., if and only if $[\alpha * \bar{e}_{b_0}] \in H$ This occurs precisely when $[\alpha] \in H$. ∎

Corollary 82.2. *The space B has a universal covering space if and only if B is path connected, locally path connected, and semilocally simply connected.*

Exercises

1. Show that a simply connected space is semilocally simply connected.

2. Let X be the infinite earring in $\mathbb{R}^2$. (See Example 1 of §80.) Let $C(X)$ be the subspace of $\mathbb{R}^3$ that is the union of all line segments joining points of $X \times 0$ to the point $p = (0, 0, 1)$. It is called the ***cone*** on X. Show that $C(X)$ is simply connected, but is not locally simply connected at the origin.

*Supplementary Exercises: Topological Properties and π_1

The results of the preceding section tell us that the appropriate hypotheses for classifying the covering spaces of B are that B is path connected, locally path connected, and semilocally simply connected. We now show that they are also the correct hypotheses for studying the relation between various topological properties of B and the fundamental group of B.

1. Let X be a space; let $\mathcal{A}$ be an open covering of X. Under what conditions does there exist an open covering $\mathcal{B}$ of X refining $\mathcal{A}$ such that for each pair B, B' of elements of $\mathcal{B}$ that have nonempty intersection, the union $B \cup B'$ lies in an element of $\mathcal{A}$?
   (a) Show that such a covering $\mathcal{B}$ exists if X is metrizable. [*Hint:* Choose $\epsilon(x)$ so $B(x, 3\epsilon(x))$ lies in an element of $\mathcal{A}$. Let $\mathcal{B}$ consist of the open sets $B(x, \epsilon(x))$.]
   (b) Show that such a covering exists if X is compact Hausdorff. [*Hint:* Let $A_1, \dots, A_n$ be a finite subcollection of $\mathcal{A}$ that covers X. Choose an open covering $C_1, \dots, C_n$ of X such that $\bar{C}_i \subset A_i$ for each i. For each nonempty subset J of $\{1, \dots, n\}$, consider the set
   $$B_J = \bigcap_{j \in J} A_j - \bigcup_{j \notin J} \bar{C}_j.]$$

2. Prove the following:
   Theorem. Let X be a space that is path connected, locally path connected, and semilocally simply connected. If X is regular with a countable basis, then $\pi_1(X, x_0)$ is countable.
   Proof. Let $\mathcal{A}$ be a covering of X by path-connected open sets A such that for each $A \in \mathcal{A}$ and each $a \in A$, the homomorphism $\pi_1(A, a) \to \pi_1(X, a)$ induced by inclusion is trivial. Let $\mathcal{B}$ be a countable open covering of X by nonempty path-connected sets that satisfies the conditions of Exercise 1. Choose a point $p(B) \in B$ for each $B \in \mathcal{B}$. For each pair B, B' of elements of $\mathcal{B}$ for which $B \cap B' \neq \varnothing$, choose a path $g(B, B')$ in $B \cup B'$ from $p(B)$ to $p(B')$. We call the path $g(B, B')$ a *select path*.

Let B_0 be a fixed element of $\mathcal{B}$; let $x_0 = p(B_0)$. Show that if f is a loop in X based at x_0, then f is path homotopic to a product of select paths, as follows:

(a) Show that there is a subdivision

$$0 = t_0 < \cdots < t_n = 1$$

of $[0, 1]$ such that f maps $[t_{n-1}, t_n]$ into B_0, and for each $i = 1, \ldots, n-1$, f maps $[t_{i-1}, t_i]$ into an element B_i of $\mathcal{B}$. Set $B_n = B_0$.

(b) Let f_i be the positive linear map of $[0, 1]$ onto $[t_{i-1}, t_i]$ followed by f. Let $g_i = g(B_{i-1}, B_i)$. Choose a path α_i in B_i from $f(t_i)$ to $p(B_i)$; if $i = 0$ or n, let α_i be the constant path at x_0. Show that

$$[f_i] * [\alpha_i] = [\alpha_{i-1}] * [g_i].$$

(c) Show that $[f] = [g_1] * \cdots * [g_n]$.

3. Let $p : E \to X$ be a covering map such that $\pi_1(X, x_0)$ is countable. Show that if X is regular with a countable basis, so is E. [*Hint:* Let $\mathcal{B}$ be a countable basis for X consisting of path-connected sets Let $\mathcal{C}$ be the collection of path components of $p^{-1}(B)$, for $B \in \mathcal{B}$. Compare Exercise 6 of §53.]

4. Prove the following:
Theorem. *Let X be a space that is path connected, locally path connected, and semilocally simply connected. If X is compact Hausdorff, then $\pi_1(X, x_0)$ is finitely generated, and hence countable.*
Proof. Repeat the proof outlined in Exercise 2, choosing $\mathcal{B}$ to be finite. One has the equation

$$[f] = [g_1] * \cdots *.[g_n],$$

as before. Choose, for each $x \in X$, a path β_x from x_0 to x; let β_{x_0} be the constant path. If $g = g(B, B')$, define

$$L(g) = \beta_x * (g * \bar{\beta}_y),$$

where $x = p(B)$ and $y = p(B')$. Show that

$$[f] = [L(g_1)] * \cdots * [L(g_n)].$$

5. Let X be the infinite earring (see Example 1 of §80). Show that X is a compact Hausdorff space with a countable basis whose fundamental group is uncountable. [*Hint:* Let $r_n : X \to C_n$ be a retraction. Given a sequence $a_1, a_2, \ldots$ of zeros and ones, show there exists a loop f in X such that, for each n, the element $(r_n)_*[f]$ is trivial if and only if $a_n = 0$.]